DATA ANALYSIS COMPETITION
INSTITUT TEKNOLOGI SEPULUH NOPEMBER

FINAL REPORT DAC 2025

FINAL STAGE 1

TITLE OF THE REPORT



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CHAPTER I: Introduction

Music, as a universal language, transcends cultural boundaries, evokes emotions, and shapes identities [1]. Within the vast landscape of musical expression, genres serve as signposts, guiding listeners through diverse sonic experiences ranging from the soothing melodies of classical compositions to the pulsating rhythms of electronic dance music. The diverse music genre stands as a fundamental endeavor in the realm of musicology and technology, fueling endeavors such as personalized music recommendations, genre-specific playlists, and thematic radio stations [2].

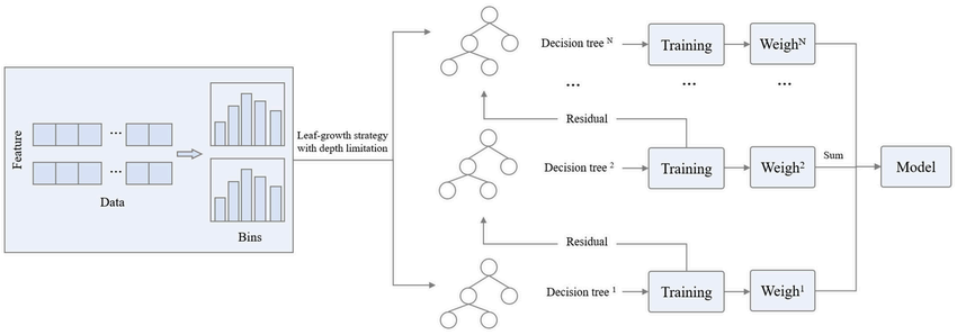
Archipelago Records, needs to analyze the rapid trend cycles around music. The massive volume of content also has rendered Archipelago's traditional business model ineffective, marked by a decline in market share and difficulties in launching successful new artists. Thus, the company faces critical challenges in efficiently allocating investment resources, identifying talent with high commercial potential, and maintaining the relevance of its genre portfolio. Since the traditional way of relying solely on qualitative methods is no longer sufficient, the application of data analytics and machine learning to music data aims to create data-driven solutions that are beneficial to working with music.

Thus, this report aims to:

a. Identify the audio and lyrical factors that significantly influence a song's popularity. Based on these findings, build and validate a regression model capable of predicting a song's popularity level.
b. Analyze and map the distinct lyrical characteristics of each music genre. The goal is to build a "lyrical profile" for each major genre.
c. Evaluate the evolution and resilience of each music genre's popularity based on its release date. Identify which genres show stable popularity trends (evergreen) and which exhibit high volatility.

CHAPTER II: Theoretical Framework

a. Light Gradient Boosting Machine (LightGBM)



Light Gradient Boosting Machine is an open-source high-performance machine learning framework developed by Microsoft [3]. It is an ensemble learning framework that uses gradient boosting method which constructs a strong learner by sequentially adding weak learners in a gradient descent manner [4].

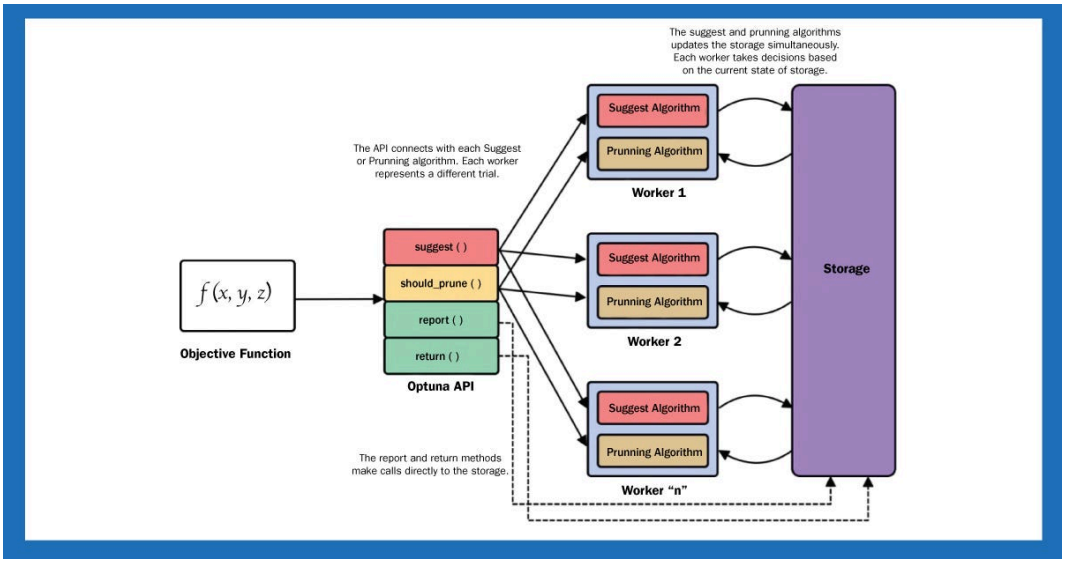
Light Gradient Boosting Machine is designed for efficiency, scalability and high accuracy particularly with large datasets. It uses decision trees that grow efficiently by minimizing memory usage and optimizing training time. Key innovations like Gradient-based One-Side Sampling (GOSS), histogram-based algorithms and leaf-wise tree growth are enabled. GOSS preserves examples with large gradients (e.g., those greater than a specific threshold or in the top percentiles), while maintaining the accuracy of information gain predictions by randomly discarding examples with small gradients [5]. This method allows for more accurate gain predictions compared to regular random sampling at the desired example ratio when information gain varies widely. It generally provides an almost lossless strategy to represent high-dimensional sparse data with fewer features. Especially in a sparse feature space, many features are mutually exclusive, meaning they do not simultaneously take non-zero values. These features can be safely merged into a single feature. As a result, the complexity of histogram construction decreases from O(data^feature) to O(data^bundle) levels (bundle < feature). [6]

LightGBM to outperform other frameworks in both speed and accuracy. Assume that the dataset have $N = \{1, 2,\ldots n\}$ and the model has $T = \{1, 2,\ldots t\}$ tree that has been produced. After iteration t times, the final prediction is the same with the count of the first $(1 - t)th$ and tth. Iteration process is illustrated in below:

$$y_i^{\wedge(t)} = y_i^{\wedge(t-1)} + f_i(x_i) \ldots (2.1)$$

Where $y_i^{\wedge(t)}$ is the score of prediction ith at iteration tth. $y_i^{\wedge(t-1)}$ shows the tree based model created before and $f_i(x_i)$ shows the new model that is just built [7]. Therefore, according to equation (1), each new prediction is generated by the residual and the previous prediction.

b. Optuna Optimization Algorithm



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Optuna is an automatic hyperparameter optimization software framework, particularly designed for machine learning [8]. It automates the search for the best hyperparameters, taking the guesswork out of tuning and allowing you to focus on developing your model. Optuna uses advanced algorithms like the Tree-structured Parzen Estimator (TPE) and CMA-ES to efficiently find optimal settings. Optuna uses Bayesian Optimization to efficiently search for the optimal hyperparameters. It starts by sampling a few sets of hyperparameters and evaluating their performance. Then, it builds a model to predict which hyperparameters might perform well based on the results so far. This model helps Optuna focus on the most promising areas of the search space, making the optimization process more efficient. The Tree-structured Parzen Estimator (TPE) is an algorithm used by Optuna for Bayesian Optimization. Instead of using a Gaussian Process like traditional Bayesian methods, TPE models the objective function using two probability density functions: one for the good hyperparameter sets and one for the others [9]. It then uses these distributions to sample new hyperparameter sets that are more likely to perform well. Below is the mathematical formula for optuna optimization:

$$\frac{l(x)}{g(x)},\ where\ l(x) = P(x|y < y^{*}),\ g(x) = P(x|y \geq y\ *) \ldots (2.2)$$

In TPE, two PDFs are constructed: $l(x)$ for good hyperparameter values and g(x) for the rest. The algorithm samples new hyperparameters by maximizing the ratio. Expected Improvement (EI) is like deciding which direction to explore next on your treasure map. It measures how much better you can expect the new hyperparameters to perform compared to your current best set. The EI for a new set of hyperparameters x is calculated as:

$$\mathrm{EI}(x) = \mathbb{E}\left[\max(y^{*} - f(x), 0)\right] \ldots (2.3)$$

where y^{*} is the best-observed value, and $f(x)$ is the predicted value of the objective function at x.

c. Augmented Dickey Fuller Test

Augmented Dickey-Fuller (ADF) test is a statistical test in time series analysis used to determine whether a given time series is stationary [10]. ADF has a function to ensure stationary across the dataset or it means consistent over time. It tests the null hypothesis that a unit root is present in a time series sample. The presence of a unit root means the series is non-stationary. Below is the mathematical function of ADF:

$$\Delta yt = \alpha + \beta t + \gamma yt - 1 + \sum i = 1p\theta i\Delta yt - i + \epsilon t \ldots (2.4)$$

where α is a constant, β the coefficient on a time trend and p the lag order of the autoregressive process. By including lags of the order p, the ADF formulation

allows for higher-order autoregressive processes. This means that the lag length p must be determined in order to use the test.

CHAPTER III: Analytical Steps

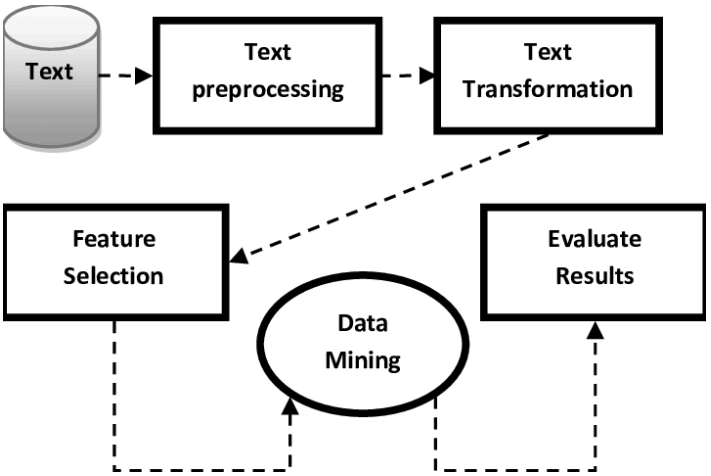


Figure 3.1. Flow Diagram

Data mining is the process of extracting or understanding interesting patterns. It is clear that text mining is a subset of data mining to extract usable information for certain purposes from a collection of documents by looking for patterns in the text. Process Stage for Text Mining, which is implemented throughout the research:

a. Text Acquisition
   Since the dataset is already provided, there's no technique used to obtain the data, for example *web scraping*. The dataset consists of 35231 rows and 19 columns in total.
b. Text Preprocessing
   Text preprocessing in NLP is the process of converting raw text into a structured format suitable for machine learning models. The goal is to clean and standardize text data to reduce noise and highlight meaningful patterns. First, there's no null and duplicated value in the dataset so we continue to the next steps. Preprocessing includes changing the abbreviated years into its complete form and removing irrelevant rows, dropping duplicate tracks, cyclical feature engineering of release_year to capture continuous year relation, use PolynomialFeatures for numeric features to capture non-linear relationship, normalize the numeric features using StandardScaler and one-hot encoded the categorical features. The dataset is then split into train and test using 0.15 as the test size. Also, for Exploratory Data Analysis (EDA), stopwords are removed.
c. Text Transformation
   Text transformation is the process of translating or converting a sequence of text to another sequence of the same or different length. This is accomplished by using sequence-to-sequence (or Seq2seq) models in the

natural language processing (NLP) domain. But, since for modelling the task is regression, it's not necessary to convert text to sequence.

d. Regression Model

The regression model uses LGBMRegressor and is fine-tuned by Optuna with 30 n_trials. The main objective of the model is to maximize the R-squared value in order to make the model more precise at predicting songs' popularity. The score is calculated by using 3-fold cross-validation.

CHAPTER IV: Analysis of Results

a. Exploratory Data Analysis (EDA)

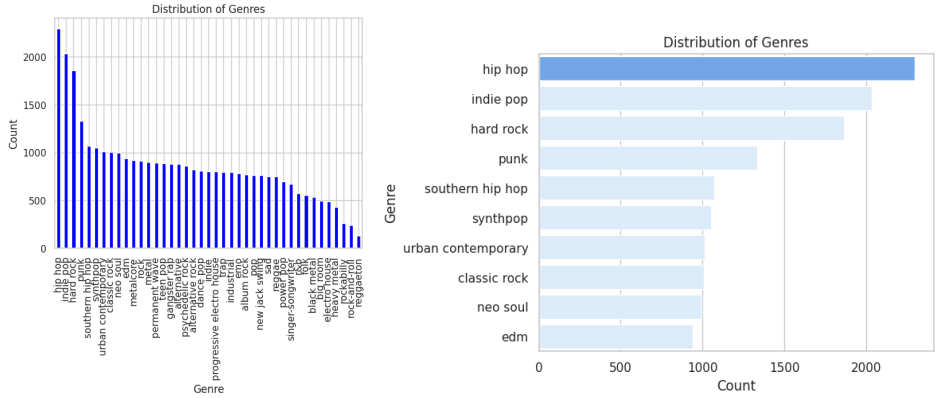


Figure 4.1. Distribution Bar Chart Based on Genre

The distribution of dataset is actually spreading quite evenly among different genres, but based on the upper bar chart 'hip hop' is the most frequent music genre in this dataset, followed by indie pop, and hard rock.



Figure 4.2. Word Cloud per Genre

Based on the word cloud visualization, each genre has different kinds of words in their lyrics. For example, even genre 'hip hop' but different types like southern hip hop and indie hiphop. As seen on the top, southern hip hop tends to have harsh and inappropriate words like '*nigga*'

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and '*shit*'. While, compared to indie pop, it just has normal words like '*love*' and '*know*'.

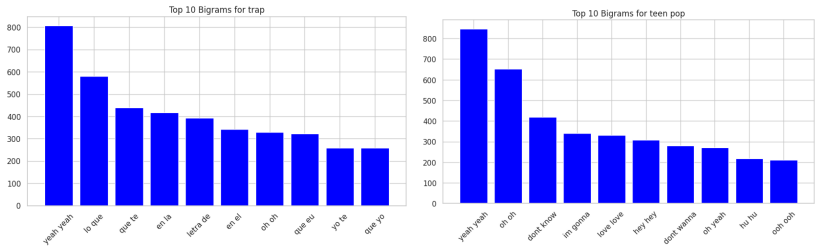


Figure 4.3. Bigrams Bar Chart per Genre

Based on the bar chart each genre has different types of bigrams. In the upper image, interestingly for trap the bigrams are not in english words. While, from teen pop the bigrams revolve around love like bigram '*love love'*.

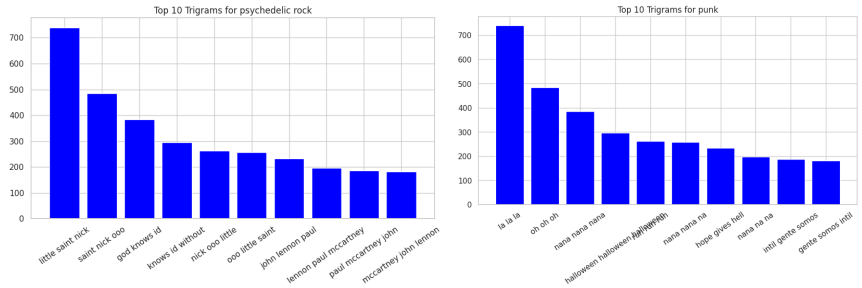


Figure 4.3. Trigrams Bar Chart per Genre

Based on the bar chart, for trigrams is actually quite unique. It's actually in the chorus or reff parts since the words are repeatable and tend to have no meaning like '*la la la*'.

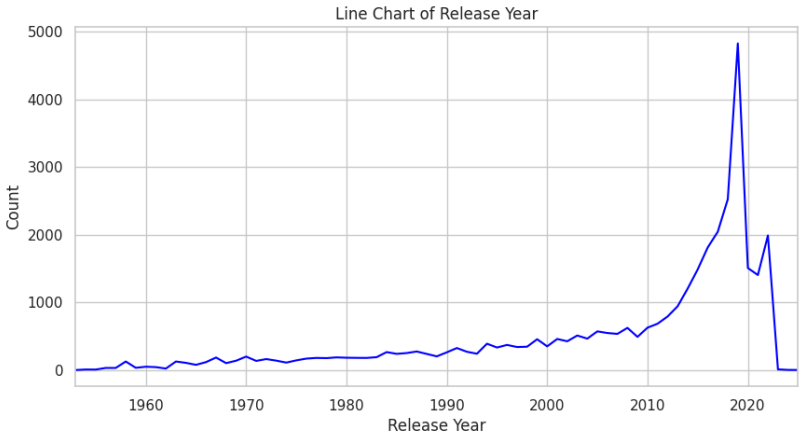


Figure 4.4. Line Chart Based on Release Year

Based on the line chart, actually most of the data is taken around 2018-2019 since there's a sudden spike in those years.

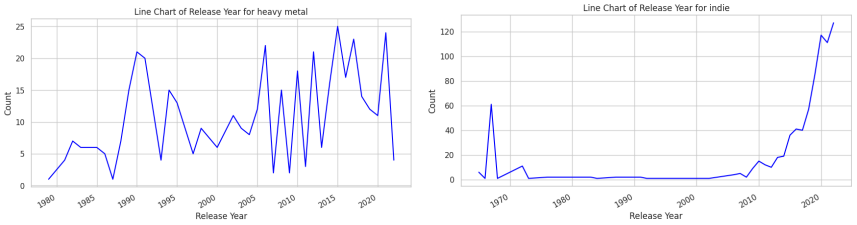


Figure 4.5. Line Chart Based per Genre

Based on the line chart, each genre has different kinds of trends. But, there's a few that are interesting. Indie is spiking up from the 2010s. And heavy metal is quite volatile.

b. Regression Model

R-squared is used as the baseline to validate the model's performance. The result shows that the model achieved the R-squared value of 0.23270. In this case, the score is considered acceptable as a song's popularity depends on human behavior which introduces tremendous variability.

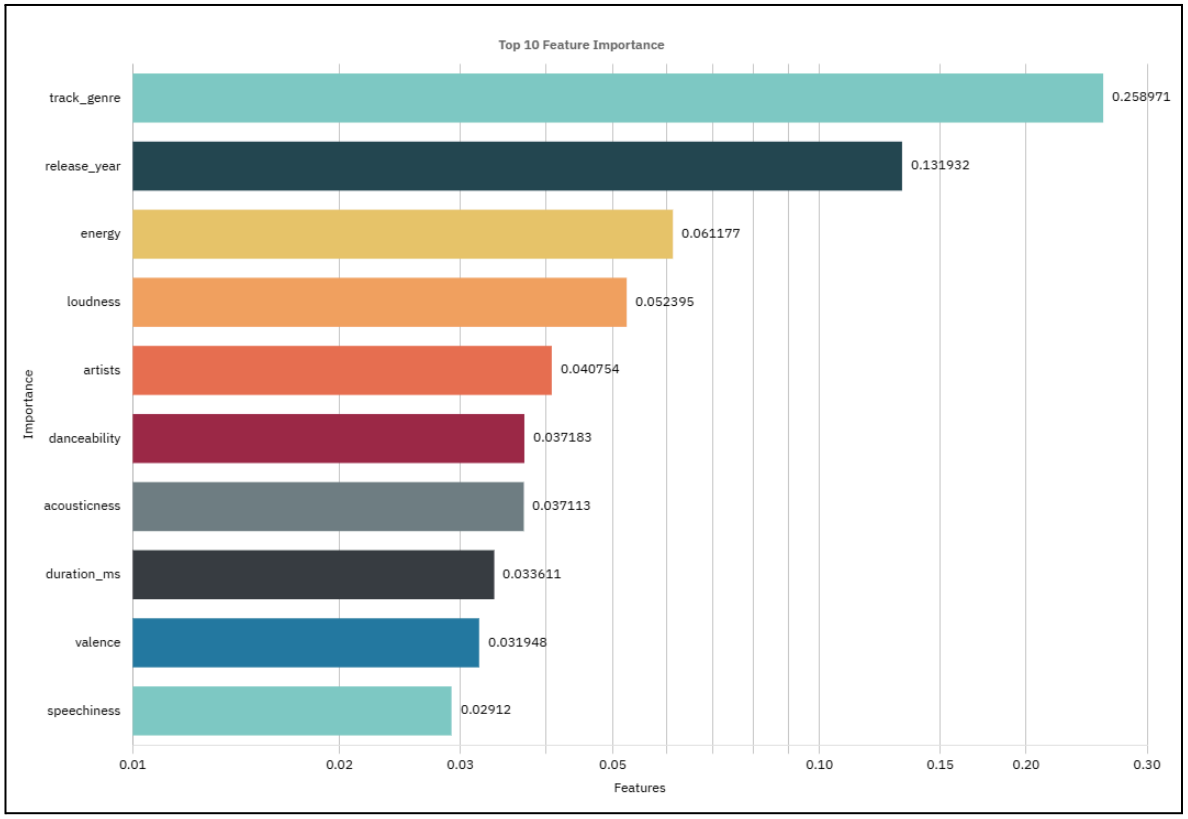


Figure 4.6. Top 10 Feature Importance

Based on the chart above, track_genre shows the most importance among other features which means that selecting the right genre is one of the key aspects in order to gain high popularity.

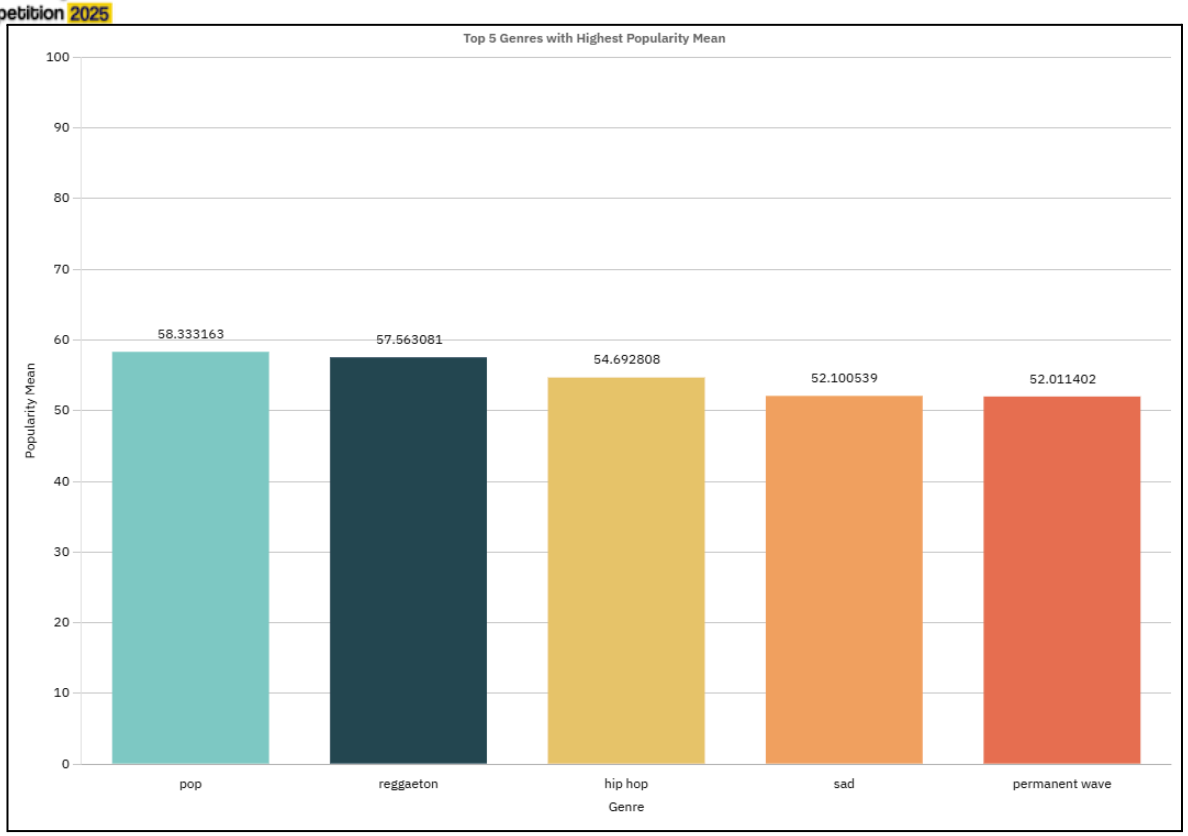


Figure 4.7. Top 5 Genres by Popularity Mean According to Model

This chart shows that pop is the genre with the highest popularity mean. In addition, other features with highest popularity mean shows that a combination of Charlie Puth or BTS, 3-minutes duration, high danceability, low energy, song with the key E, moderate loudness, a balance between music and speech, low acousticness, high tempo and a well-structured lyrics is a good track combination according to the model.

c. Result of Dickey-Fuller Test

Table 4.1. Result of Dickey-Fuller Test

Genre	p-value
hip hop	0.4414
southern hip hop	0.2425
album rock	0.0109
classic rock	0.1542
permanent wave	0.2502
urban contemporary	0.9983
new jack swing	0.9647
edm	0.5783

progressive electro house	0.7162
folk	0.0199
metalcore	0.97647
pop	0.9973

Based on a few genre Dickey-Fuller tests, we can conclude that there are a few potential genres that are consistent whenever the time period is (or called stationary). If the p-value is lower than 0,05, it means that we reject the H_0 and accept the H_1 that the genre based on the time frame is indeed stationary. The potential one is album rock and edm based on the upper chart.

CHAPTER V: Conclusion and Recommendation

The conclusion that can be achieved is that every genre has its own unique selling point and each has distinct characteristics of words that are used. From the bar chart visualization, as can be seen, the most popular genre right now is in 'pop', where singers like 'BTS' and 'Charlie Puth' dominate. Whileas, from the time series analysis, the genre that is quite interesting right now is 'indie' since there is a spike in recent years. But, unfortunately, as we built the regression model, the model didn't really capture and can predict the patterns that actually make the song go booming or famous since the R-squared value only achieved around 0.23270 eventhough it has been optimized.

Recommendation for next steps is that the company hopefully can increase its singer in pop genre since it has been proven to be the most famous one in these days. Also, indie might be an option too since it's gaining popularity since the last years. Lastly, for predictive modelling, the company can try a larger and complex model like neural networks that can capture more meaningful pattern.

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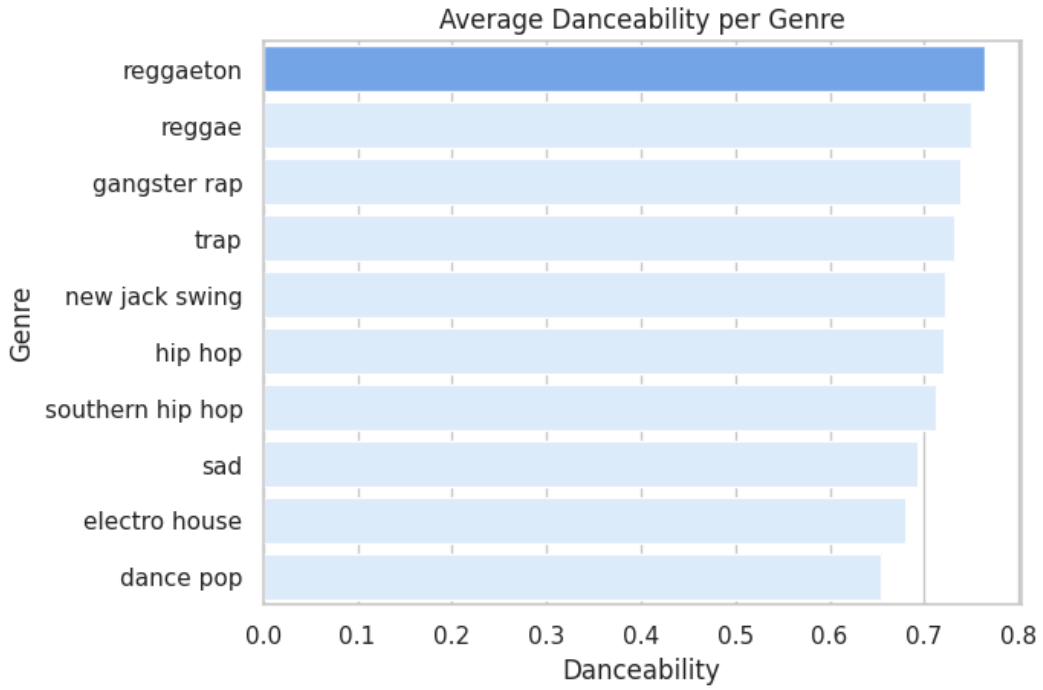
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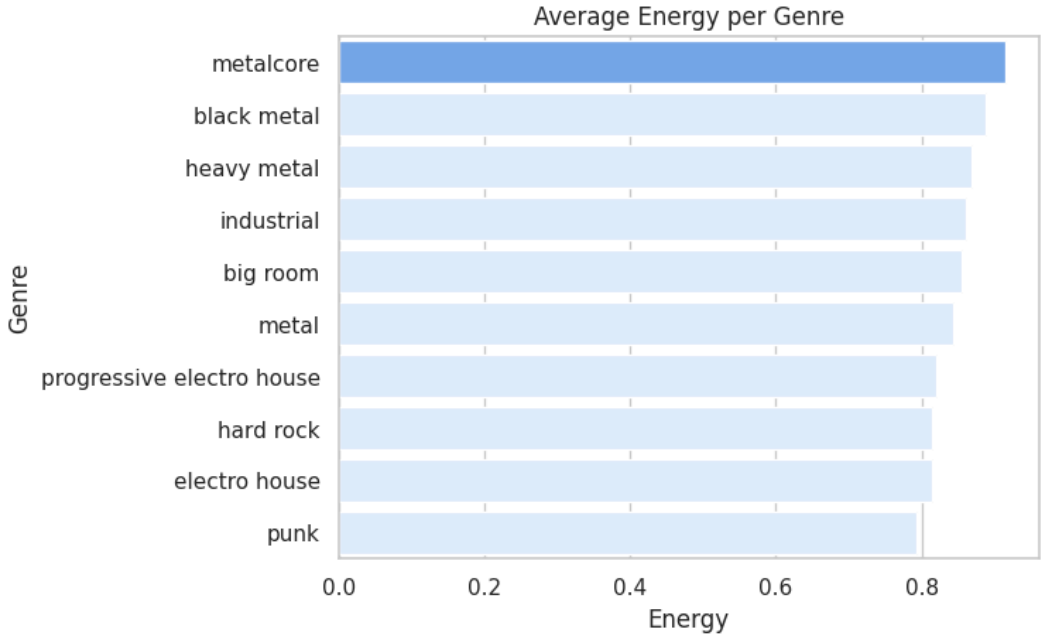
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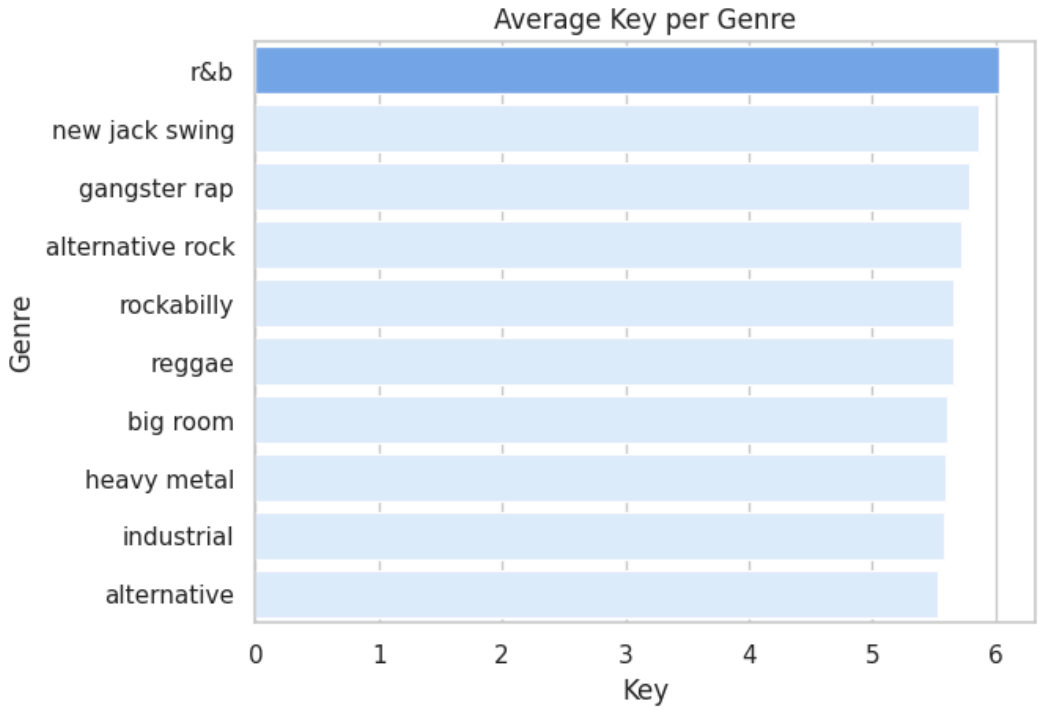
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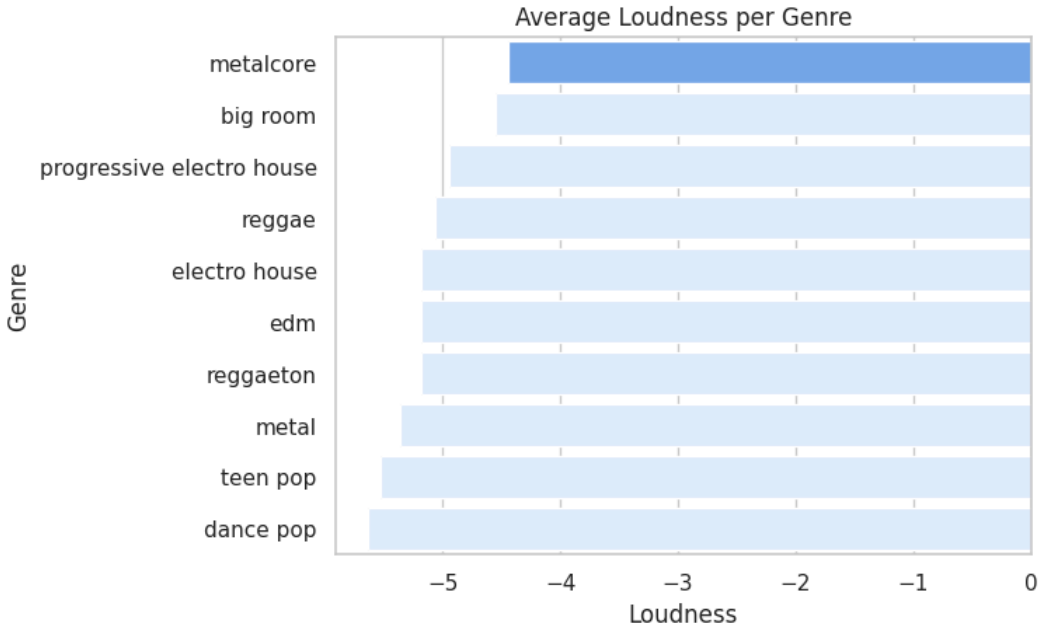
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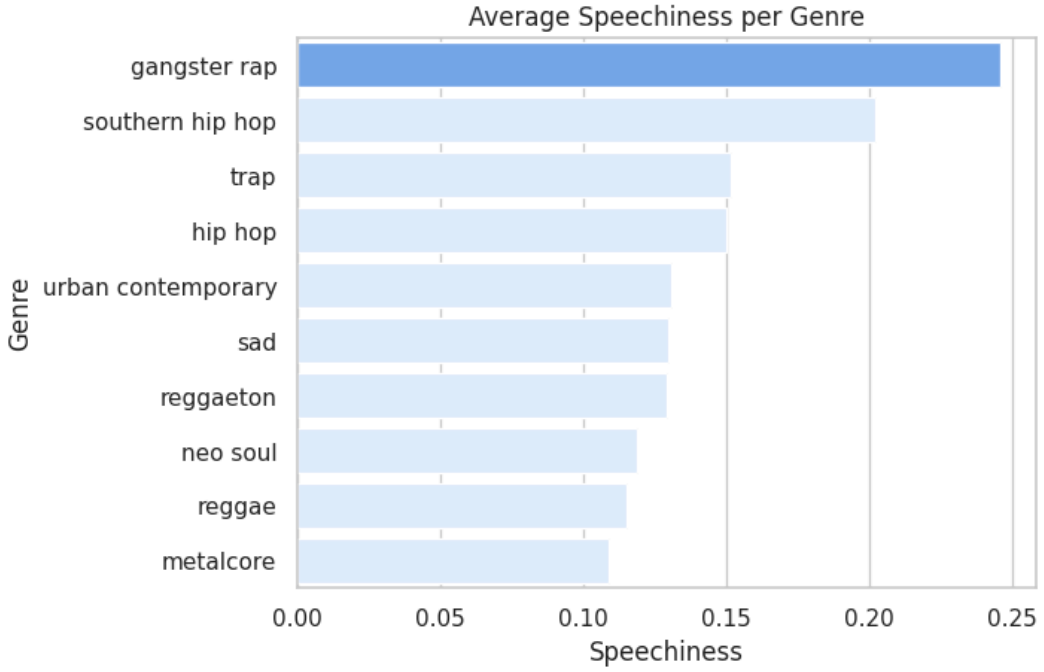


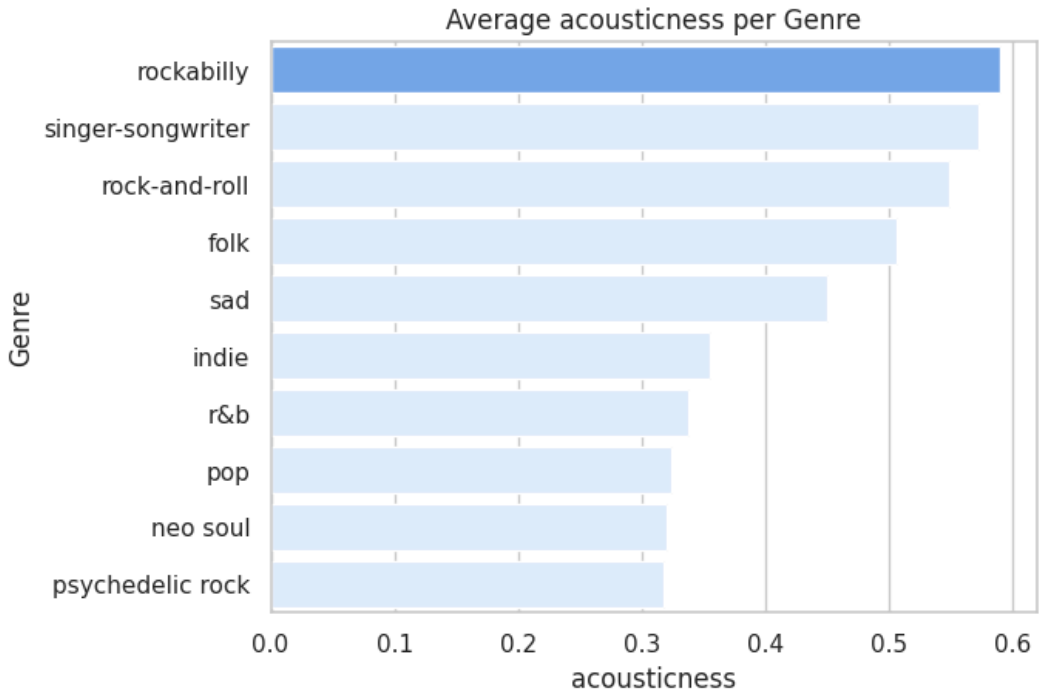
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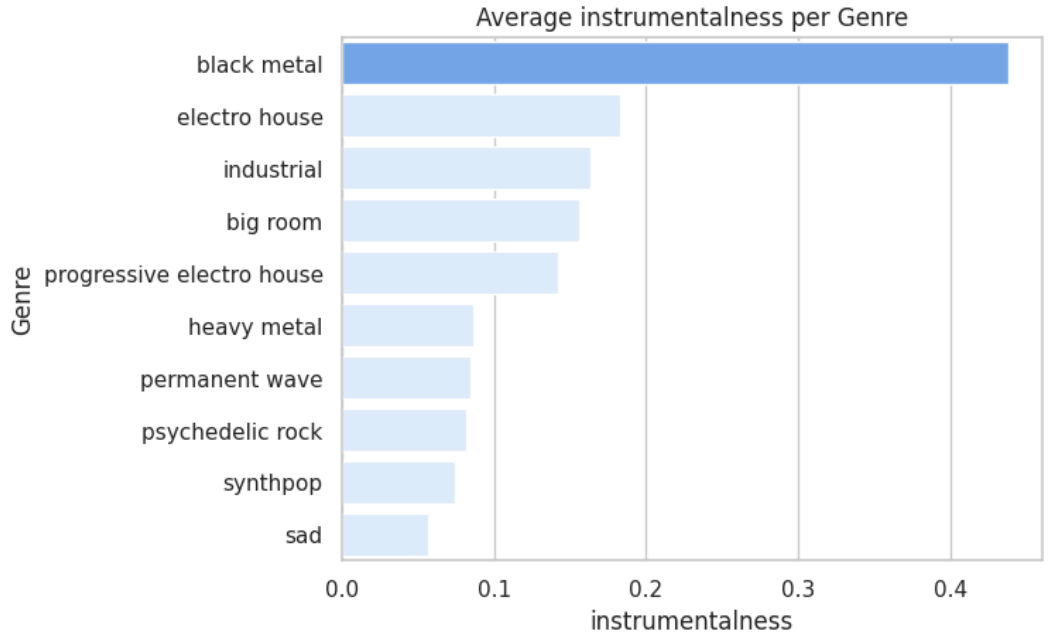


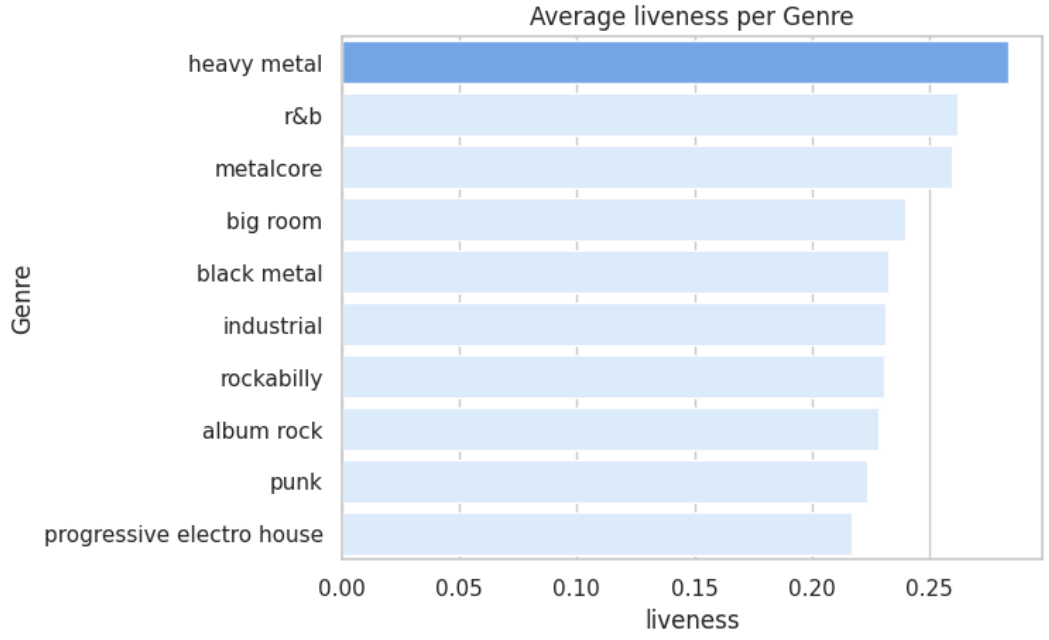
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